BrightLight Tutorials – Data Analytics

SQL Fundamentals

Exercise 3: SQL CASE Statements

Instructions:

1. Write your answers on paper using a pen.
2. For each query, **draw a table** showing the final output (the result set).
3. In your `SELECT` statements, choose **relevant columns to display**, unless specified.
4. After completing all questions, **scan your work into a PDF**.
5. Email your PDF to: **rofhiwa@brightlighttutorials.co.za**
6. **Submission Deadline: 21 April 2025, 23:59**

Questions

1. Table: `products`

product_id	product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	Keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

Syntex

```
Select product_name,
       price,
Case
When price > 1000
Then 'Expensive
When price between 100 And 1000
Then 'Mid-rang'
Else 'Budge'
End AS price_Category
from products;
```

Question:

Classify each product by price:

- 'Expensive' if price > 1000
- 'Mid-range' if price between 100 and 1000
- 'Budget' if price < 100

Expected Output Columns:

- `product_name`
- `price`
- `price_category`

Out Put

Product_name	Price	Price_category
Laptop	1200	Expensive
Phone	800	Mid-range
Keyboard	45,00	Budget
Monitor	300	Mid-range
Mouse	25,00	Budget

2. Table: orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

Sdntex

```
Select Customer_name, amount
Case
When amount >= 1000 Then 'High Value'
When amount between 500 And 999,99
Then 'Medium Value'
Else 'Low Value'
End AS Order_Value_Category
from Orders;
```

Question:

Label each order:

- 'High Value' for orders ≥ 1000
- 'Medium Value' for 500–999.99
- 'Low Value' for orders < 500

Expected Output Columns:

- `customer_name`
- `amount`
- `order_value_category`

Out Put

Customer_name	amount	order_Value_Category
Alice	150	Low Value
Bob	560	Medium Value
Charlie	999,99	Medium Value
Diana	45,50	Low Value
Ethan	1200	High Value

3. Table: `employees`

emp_id	emp_name	department	salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

Question:

Categorize employee position:

- If in 'IT' and salary > 80000 → 'Senior IT'
- If in 'HR' and salary > 55000 → 'Experienced HR'
- Otherwise → 'Staff'

Expected Output Columns:

- `emp_name`
- `department`
- `salary`
- `position_level`

Syntex

```
Select emp_name, department,
 Salary
Case
When department = 'IT' And
Salary > 80 000 Then
'Senior IT'
When department = 'HR' And
Salary > 55000 Then
'Experienced HR'
Else 'Staff
End AS position_level
From employees;
```

Output

emp_name	department	Salary	Position_level
John	IT	85000	Senior IT
Sara	HR	60000	HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	Staff

Syntex

4. Table: students

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

```
Select Student_name, Score
Case
When score >= 90 Then 'A'
When Score between 80 And 89
Then 'B'
When Score between 70 And 79
Then 'C'
When Score between 60 And 69
Then 'D'
Else 'F'
End AS grade
From Students;
```

Question:

Assign a letter grade:

- ≥ 90: 'A'
- 80–89: 'B'
- 70–79: 'C'
- 60–69: 'D'
- < 60: 'F'

Expected Output Columns:

- `student_name`
- `score`
- `grade`

OutPut

Student-name	Score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

5. Table: `deliveries`

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

Syntex

```
Select delivery_id, delivery_time_minutes
case
When delivery_time_minutes <= 30
Then 'Fast'
When delivery_time_minutes between
31 And 60 Then 'On Time'
Else 'Late'
End As Performance
From deliveries;
```

Question:

Label delivery performance:

- ≤ 30 mins: 'Fast'
- 31–60 mins: 'On Time'
- 60 mins: 'Late'

Expected Output Columns:

- `delivery_id`
- `delivery_time_minutes`
- `performance`

output

delivery_id	delivery_time_minutes	performance
1	45	On Time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

Syntex

```
Select issue-type, priority,
Case
When priority = 3 Then 'High'
When priority = 2 Then 'Medium'
Else 'Low'
End AS priority_label
From tickets;
```

6. Table: `tickets`

ticket_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	Slow system	2
4	Email error	2
5	Password reset	1

Question:

Convert priority to labels:

- 3 → 'High'
- 2 → 'Medium'
- 1 → 'Low'

Expected Output Columns:

- `issue_type`
- `priority`
- `priority_label`

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow system	2	Medium
Email error	2	Medium
Password reset	1	Low

7. Table: `attendance`

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

Question:

Calculate attendance % and classify:

- ≥ 90% → 'Excellent'
- 75–89% → 'Good'
- < 75% → 'Needs Improvement'

Expected Output Columns:

- `student_id`
- `attendance_percentage`
- `attendance_status`

Syntex

```
Select Student_id, days_Present,
total_days,
('days_present'/total_days)*100
AS attendence_percentage,

Case
When('days_present'/total_days)*100
>= 90 Then 'Excellent'
When('days_present'/total_days)*100
between 25 and 80 Then 'Good'
Else 'Need_Improvement'
End AS attendance_status
From attendance;
```

Out Put

Student_id	total_days	days_present	attedance_percetage
1	50	45	90
2	50	30	60
3	50	48	96
4	50	25	50
5	50	50	100

Syntex

8. Table: `products_inventory`

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

Question:

Label stock status:

- 0 → 'Out of Stock'
- 1–5 → 'Low Stock'
- 5 → 'In Stock'

Expected Output Columns:

- `product_id`
- `stock_qty`
- `stock_status`

```
Select product_id; Stock_qty,
Case
When Stock_qty = 0 Then 'Out of Stock'
When Stock_qty between 1 And 5
Then 'Low Stock'
Else 'In Stock'
End AS Stock_Status
From product_inventory;
```

Out Put

Product_id	Stock_qty	Stock_status
1	5	Low Stock
2	0	Out of Stock
3	25	In Stock
4	10	In Stock
5	3	Low Stock

Syntex

9. Table: `classes`

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

```
Select Subject, enrolled_Students
Case
When enrolled_Students >= 25
Then 'Large
When enrolled_Students between
10 And 25 Then 'Medium'
Else 'Small'
End AS class_Size_Category
from Classes;
```

Question:

Classify by size:

- ≥ 25 → 'Large'
- 10–24 → 'Medium'
- < 10 → 'Small'

Expected Output Columns:

- `subject`
- `enrolled_students`
- `class_size_category`

Output

Subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	medium
Art	5	Small
History	20	Medium

Syntex

Select payment_id, Payment_method, amount

Case
When payment_method = Cash AND amount >= 200
Then 'Eligible for Discount'
Else 'Not Eligible'
End AS discount_eligibility
from payments;

10. Table: payments

payment_id	amount	payment_method
1	50.00	Card
2	200.00	Cash
3	150.00	Card
4	75.00	PayPal
5	300.00	Cash

Question:

Apply discount flag:

- If `payment_method = 'Cash'` and amount ≥ 200 → 'Eligible for Discount'
- Otherwise → 'Not Eligible'

Expected Output Columns:

- `payment_id`
- `payment_method`
- `amount`
- `discount_eligibility`

Payment_id	Payment_method	amount	discount_eligibility
1	Card	50,00	Not Eligible
2	Cash	200,00	Eligible for Discount
3	Card	150,00	Not Eligible
4	PayPal	75,00	Not Eligible
5	Cash	300,00	Eligible for Discount